

Computational Design and Tolerance-Tiered Optimization of Chord-Skip Toroidal Multipass Cells for Airborne Methane Spectroscopy

Abstract

Toroidal multipass cells fold long absorption paths into compact, mechanically robust volumes, but published designs demonstrate at most ~ 10 m of optical path in the 100–200 mm diameter class and typically require custom diamond-turned optics. We present a family of ring-geometry multipass cells that reach a verified 20.4–29.0 m optical path length within a 183 mm assembly envelope — and 20.7 m within 141 mm — using 12–16 catalogue 25.4 mm concave mirrors ($\approx$ US\$1000 of optics) and a sub-kilogram optical head. Three design principles enable this: (i) a chord-skip ring geometry whose near-diameter chords deliver up to 143 mm of path per reflection at $7\text{--}13^\circ$ incidence; (ii) an engineered dual-plane re-entrance condition in which the machined ring radius acts as the closure-tuning parameter (≈ 1.2 rad of accumulated transverse phase per millimetre), with a number-theoretic constellation rule that guarantees fringe-safe spot separation on every mirror; and (iii) mode-matched injection through a single 1.3 mm coupling hole, which makes hole losses negligible so that cell transmission equals R to the power $(n-1)$ in mirror reflectivity R — 86.7 % for the headline 20.4 m design at $R = 0.999$. Designs are verified with an exact three-dimensional ray tracer including astigmatic Gaussian-beam propagation, cross-validated against an independent ray tracer to sub-micrometre agreement, and toleranced by Monte-Carlo analysis: at research-grade build tolerances the headline design completes all 144 traversals in 100 % of trials, the ± 1 % catalogue tolerance on mirror focal length is absorbed by a linear 0.72 mm-per-percent trim of the ring radius, and a plain aluminium ring holds alignment over ± 26 K. Because the mirrors provide $R = 0.999$, the photon budget admits ~ 700 reflections and the true limit is spot-pattern geometry under build error; a tolerance-tiered menu results, with every tier verified by per-trial physical criteria (path completion, spot merging, hole leakage, unassisted exit): 20.4 m at 86.7 % is robust at standard laboratory build tolerances; 29.0 m at 76.6 % ($\varnothing 183$ mm), 24.8 m at 83.9 %, and 20.7 m at 81.6 % in a $\varnothing 141$ mm envelope are robust at flight-grade (0.1 mrad) build tolerances — the glued/welded construction class a vibrating airframe requires regardless; and the geometric ceiling — 38.6 m in $\varnothing 169$ mm at 72.5 % — is reachable only with active alignment.

1. Introduction

Tunable diode laser absorption spectroscopy (TDLAS) of the methane $2\nu_3$ band near 1654 nm requires tens of metres of absorption path for sub-ppm sensitivity, while drone deployment restricts the sensor head to roughly a 19 cm envelope and about a kilogram of payload. Multipass cells reconcile these demands by folding the path between mirrors. The classical two-mirror Herriott cell [11–13] achieves this with re-entrant elliptical patterns; its densest variants reach 54.6 m in a 12 cm-base astigmatic-type cell [9] and 26.4 m in a 12 cm dense-pattern cell [10], at the cost of custom mirror pairs and critical two-mirror alignment. Ring-geometry (toroidal/circular) cells trade some packing density for one-piece mechanical robustness: demonstrated paths are 2.2–4.1 m in the monolithic toroidal cell of Tuzson et al. [1, 2], >12 m in a 145 mm circular paraboloid [3], 10 m at <200 g in the segmented cell of Graf et al. [4], and 10 m (30 m theoretical) from multi-layer patterns on a 100 mm ring [7]; mask-based fringe suppression [5], double-row geometries at larger diameter [6], and cryogenic toroidal cavities for particle-physics spectroscopy [8] extend the family. Across this literature, demonstrated toroidal-class paths in drone-compatible envelopes cluster at or below ~ 10 m, and every design relies on custom optics.

This work asks a deliberately constrained question: how much verified optical path can a ring cell deliver inside a 190 mm envelope using only catalogue mirrors — Thorlabs CM254-series 25.4 mm concave mirrors (16 focal lengths, ≈US\$74 each) — with 9–16 mirrors, a single 1.3 mm coupling hole, and a 1.3 mm collimated input? The chord-skip parameter is the entry point: in our early design-space exploration, moving from adjacent-mirror propagation to a skip of $s = 4$ on a 9-mirror ring raised the path length by 67 % while cutting the incidence angle from 67.5° to 10° , and a companion catalogue study showed 12.7 mm mirrors saturate near 12 m — fixing the present 25.4 mm design space. Our contributions are: (1) the chord-skip parametrisation itself, with $s \approx N/2$ giving near-diameter chords at near-normal incidence; (2) an engineered — rather than searched-for — re-entrance condition, tuned by the machined ring radius against catalogue-locked curvatures, with an exact constellation criterion guaranteeing fringe-safe spot patterns before any ray is traced; (3) mode-matched single-hole coupling that renders the hole lossless; and (4) a tolerance-tiered design menu (13.6–38.6 m), in which every design passes an eight-item physical check matrix in exact ray tracing, is cross-validated independently, and carries a Monte-Carlo robustness verdict at a stated build grade, together with machining-level build specifications and a drone mass budget.

2. Cell architecture and design principles

2.1 Chord-skip ring geometry

N identical concave mirrors of curvature radius R and clear-aperture radius 11.4 mm face inward from a ring of radius R_{ring} , one every $360^\circ/N$. A beam injected through a hole in mirror M0 advances by s mirrors per reflection (the chord skip, $\gcd(N, s) = 1$). Each chord length and the (identical) incidence angle on every mirror follow

$$L = 2 R_{ring} \sin(\pi s/N), \quad \theta_i = \pi/2 - \pi s/N. \quad (1)$$

For $s \approx N/2$ the chords approach the ring diameter while θ_i falls to 6 – 13° : a 72 mm-radius ring yields 140+ mm of path per reflection — the geometric core of the result. The assembly envelope is $2(R_{ring} + 18 \text{ mm})$, the allowance covering substrate and housing wall.

2.2 Re-entrance: dual-plane phase closure

In the paraxial unit cell (one chord plus one mirror) the transverse ray oscillation advances per bounce by a phase θ in each principal plane, split by the off-axis astigmatism:

$$\cos \theta_t = 1 - L/(R \cos \theta_i), \quad \cos \theta_s = 1 - L \cos \theta_i / R, \quad (2)$$

with per-plane stability requiring $|\cos \theta| \leq 1$. The beam exits through the entrance hole after $n = kN$ reflections when both accumulated phases return to their launch value; zero-crossing launches close at phase 0 or π , giving four closure-mode combinations per geometry:

$$n \theta_{t,s} \equiv 0 \text{ or } \pi \pmod{2\pi}, \quad n = kN. \quad (3)$$

The single machinable parameter R_{ring} moves both accumulated phases along a nearly fixed direction, at ≈ 1.2 rad per millimetre for the designs below, so a candidate is closable only if its signed phase residuals lie near that line; the residual is absorbed by the launch amplitudes through their weak aberration-mediated phase pull. Re-entrance thus becomes an engineered property: the ring is machined to the radius that closes the pattern for the delivered mirror lot, and the verified designs re-cross the hole centre to within 10^{-6} – 10^{-4} mm.

2.3 Constellation rule: fringe-safe spot patterns

At closure the n spots sample a Lissajous curve at the phase-snapped rates θ' of Eq. (3); spot j lands on mirror $(js) \bmod N$ at face coordinates

$$u_j = A_t \sin(j \theta'_t), \quad v_j = A_s \{\sin, \cos\}(j \theta'_s). \quad (4)$$

Three exact geometric rules follow: the spots-per-mirror count k must be odd (for even k the tangential π -slot coincides with a sagittal near-return and a spot lands in the coupling hole); $\gcd(M, k) = 1$ in both planes with $M = \text{round}(n\theta/2\pi)$ (otherwise the k spots per mirror collapse into clusters); and — decisive in practice — every mirror carries the same k -point pattern at a different phase origin along the curve, so the worst spot pair must be evaluated over all N mirrors, not only the launch mirror, which overestimates usable separation by up to $3\times$. The minimum pair distance and pattern extent are computed exactly per candidate from Eq. (4), sizing the required amplitude — with every clearance additionally padded by the expected build-error spot walk of the target build grade (§5) — and rejecting self-crowding patterns before ray tracing. Spot-overlap fringes, the dominant practical noise source in this cell class [5, 7], are thereby excluded by construction.

2.4 Mode-matched single-hole coupling

The cell eigenmode has a mirror-plane $1/e^2$ radius per plane of

$$w_m^2 = M^2 \lambda L / (\pi \sin \theta), \quad (5)$$

$\approx 0.3\text{--}0.45$ mm here. A 1.3 mm waist placed at the hole is badly mismatched to it; the beam then breathes to $\sim 3\times$ its injected size, spots overlap, and the hole clips 13.5 % of the power at entry and again at exit. Instead, the injection optics place the in-cell waist (0.20–0.33 mm) near mid-chord: the beam rides the eigenmode at 0.21–0.42 mm for the entire path, the radius at the hole is $\sim 0.3\text{--}0.37$ mm, and the 1.3 mm hole transmits ≈ 100 % both ways. Cell transmission is then exactly

$$T(R) = T_h^2 R^{n-1} \approx R^{n-1}, \quad (6)$$

with $T_h \approx 1$ the hole transmission. At closure the round-trip ABCD map is the identity, so the exit beam reproduces the injected Gaussian and leaves through the same hole, separated from the input axis by $2\theta_i$ (22.5° for $N = 16$), placing laser and detector side by side behind M0 with no beamsplitter.

2.5 Injection chain: source and collimator selection

The mode-matched launch is a finite-conjugate imaging task — relay the source mode to a 0.20–0.33 mm waist roughly 45–120 mm past the coupling hole — and the collimator choice follows from it. The baseline chain is a fibre-pigtailed 1654 nm DFB: the fibre mode (radius $\approx 5.5 \mu\text{m}$) is imaged by a short-focal aspheric lens of the standard collimation-package type [14] operated slightly defocused, at magnification $\approx 60\times$. For the headline design an $f \approx 2$ mm asphere places the 0.33 mm waist ≈ 120 mm from the lens — the exact conjugates, computed by the q-parameter design, are listed on each specification sheet — so a single catalogue lens in an adjustable FC/APC collimator body performs collimation and mode matching in one element, and its focus adjustment is the fine trim used during alignment. An attractive alternative first stage is a reflective (off-axis-paraboloid) fibre collimator: being achromatic and figure-stable it removes lens focal drift from the thermal budget of §5, complementing the ± 26 K window of the aluminium ring. Direct free-space collimation of a TO-can device is the most compact option but is quantitatively limited: the best residual divergence of a collimated TO60 package is $\approx 0.8^\circ$ full angle (≈ 7 mrad half-angle), whereas the cell eigenmode accepts $\lambda/(\pi w_0) \approx 1.6\text{--}2.6$ mrad — a $\approx 4\times$ mismatch, equivalent to an M^2 well beyond the verified

1.3 budget, costing $\geq 80\%$ of the light and destroying the spot-size guarantees. TO-can sources are therefore suitable only with additional spatial filtering (or for much shorter cells); the fibre-coupled variant is the design baseline.

3. Numerical methods

Designs were generated and verified with our open simulation platform: an exact 3-D ray tracer on toroidal/spherical surfaces (analytic normals, Newton-iterated intersections, per-bounce aperture tests), astigmatic Gaussian propagation via per-plane complex-q ABCD transfer with the actual per-bounce incidence angles and an M^2 factor, loss budget, per-plane stability, and spot diagnostics; earlier iterations of the platform added Sobol design-space sampling and machine-learning surrogates used in the exploratory phase. The search runs in two stages: Stage A enumerates the catalogue space ($16 \text{ curvatures} \times N = 9\text{--}16 \times \text{coprime skips} \times \text{a fine ring-radius grid} \times \text{odd } k \text{ up to } 45$) against packing, envelope, stability, closure-residual and constellation criteria, with all clearances sized for the expected build-error walk; Stage B ray-traces survivors at a design-basis $M^2 = 1.1$, sweeps the ring radius in $20 \mu\text{m}$ steps to locate the $\sim 20 \mu\text{m}$ -wide closure valley, and polishes six launch/geometry parameters by Nelder-Mead. A design is accepted only if the final trace passes all eight checks of Table 3. The tracer was cross-validated against the independent open-source ray tracer Optiland [15]: over 64 bounces of the headline design the two agree to $0.000 \mu\text{m}$ RMS in spot position, 0.01° in incidence angle, and $0.8 \mu\text{m}$ in chord length.

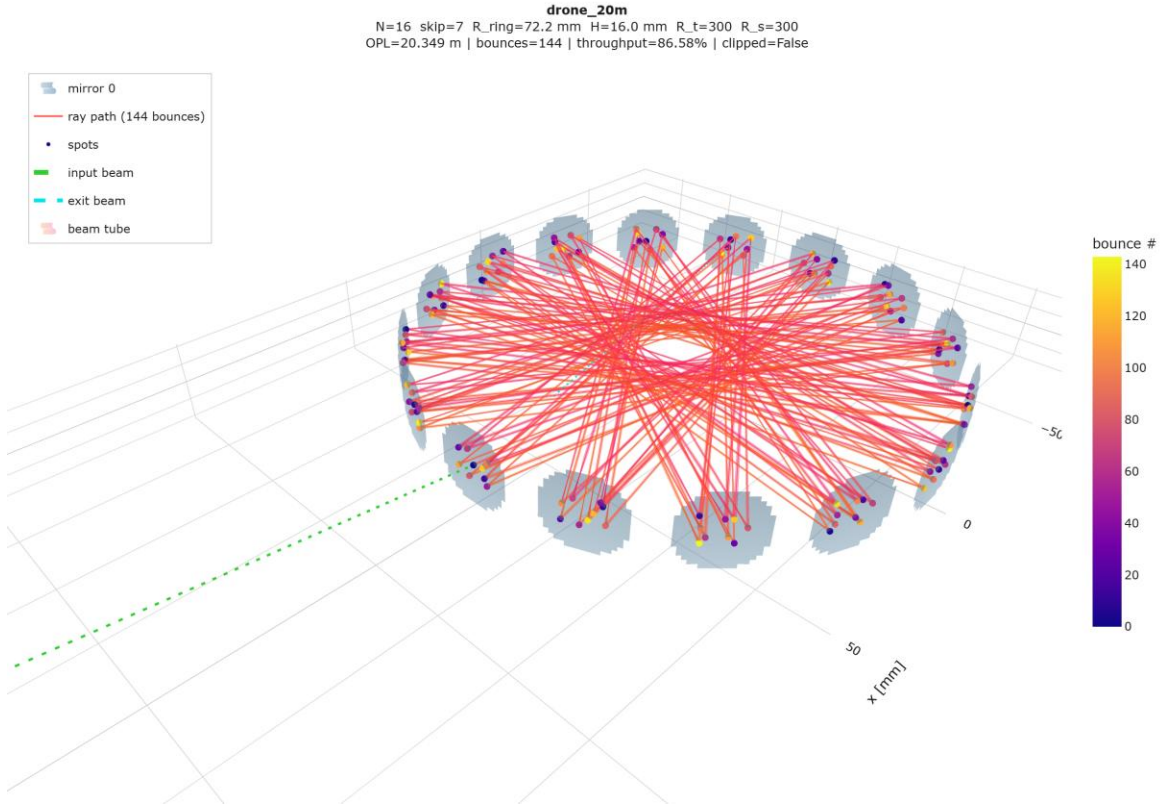


Fig. 1. Exact traced beam path of the headline design: 16 catalogue mirrors on a 72.155 mm ring, 144 chords (20.38 m), coloured by bounce order; the input beam (green) enters through the 1.3 mm hole in M0 and the closed pattern exits through the same hole.

4. Results

4.1 The tolerance-tiered design menu

Table 1. Tolerance-tiered design menu. Every row passes the full check matrix in the exact ray trace; the robustness tier is a 100-trial Monte-Carlo verdict at the stated build grade (§5). Transmission $T = R^{(\text{chords}-1)}$ exactly (hole losses ≈ 0), quoted at the project mirrors' $R = 0.999$. Envelope = $2(R_{\text{ring}} + 18 \text{ mm})$.

design	mirrors	chords	OPL [m]	T @0.999	env. $\times$ height [mm]	robust at
geometric ceiling	14 $\times$ CM254-100 (R 200)	322	38.60	72.5 %	169 $\times$ 24	active align.
flight max OPL	12 $\times$ CM254-750 (R 1500)	204	28.99	76.6 %	183 $\times$ 23	flight build
long	16 $\times$ CM254-200 (R 400)	176	24.77	83.9 %	180 $\times$ 22	flight build
mid	14 $\times$ CM254-100 (R 200)	182	22.25	83.4 %	172 $\times$ 26	active align.
flight compact	12 $\times$ CM254-500 (R 1000)	204	20.66	81.6 %	141 $\times$ 27	flight build
headline 20 m	16 $\times$ CM254-150 (R 300)	144	20.38	86.7 %	180 $\times$ 16	standard build
balanced	13 $\times$ CM254-150 (R 300)	143	16.60	86.8 %	160 $\times$ 20	flight build
small	10 $\times$ CM254-150 (R 300)	190	14.85	82.8 %	133 $\times$ 19	flight build
small max-T	12 $\times$ CM254-250 (R 500)	132	13.64	87.7 %	143 $\times$ 16	flight build

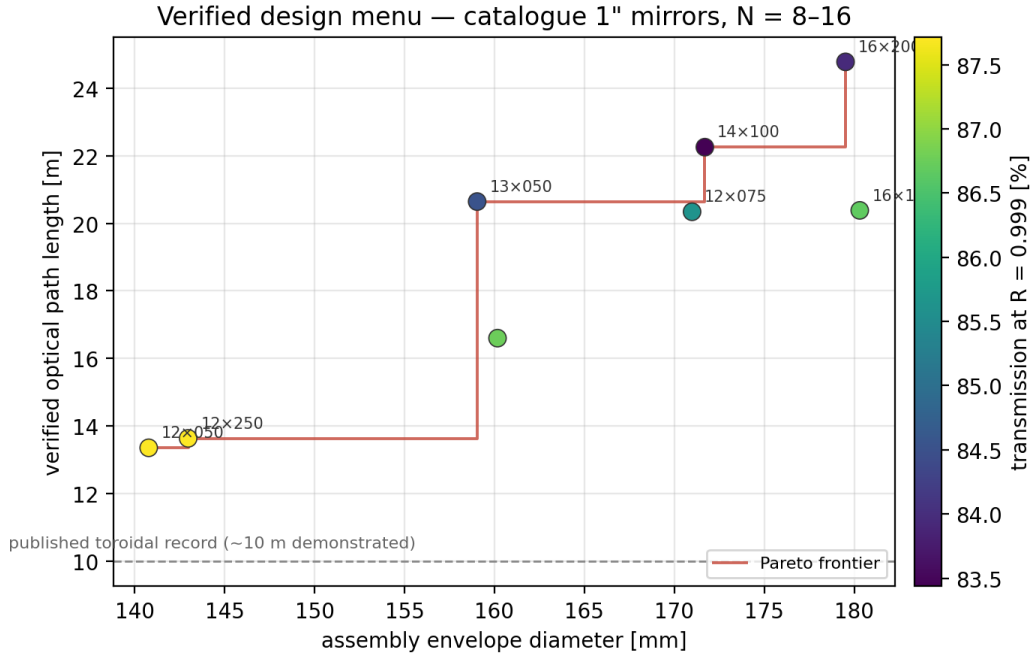


Fig. 2. The verified design menu as an envelope–path map (marker colour = transmission at $R = 0.999$). The Pareto frontier is populated entirely by catalogue-mirror designs; the dashed line marks the ~ 10 m demonstrated record of published toroidal cells in this size class.

The mirrors used in this project provide $R = 0.999$ at 1654 nm, so the photon budget admits ~ 700 reflections before transmission falls below 50 % and the real limit becomes spot-pattern geometry under build error. Table 1 therefore reports each design's robustness tier, determined by a 100-trial Monte-Carlo at the stated build grade with per-trial physical criteria (full path completion, no spot merging by the simulator's own minimum-separation metric, positive hole clearance, and an exit that still leaves the 1.3 mm hole with no

realignment): 'standard build' = kinematic-mount laboratory tolerances (0.5 mrad tilt class); 'flight build' = glued/welded precision tolerances (0.1 mrad class — the vibration spec a drone cell needs anyway); 'active alignment' = nominally feasible but requiring in-situ trim (ring temperature + launch adjuster). (Transmission for any other coating follows from Eq. (6), Fig. 7 — e.g. a protected-gold build at $R = 0.985$ would transmit 6–14 %.) The tier boundaries, established by exhaustive search (0.1 mm ring-radius grid, up to 720 bounces, $k \leq 45$ spots per mirror, clearances sized to the expected build-error spot walk), are themselves the key observation. At standard tolerances the dense long-path constellations lose spot identity — relative spot motion of 0.3–0.5 mm collapses their separation margins (the 38.6 m design's 5th-percentile minimum separation falls to 0.06 mm) — while at flight grade the 176–204-chord designs hold every margin with $\geq 99\%$ path completion. Because pattern walk grows roughly with the square root of the chord count, the standard-tolerance ceiling sits near 145 chords on the 11.4 mm clear aperture. Downward, the smallest robust envelope is 133 mm (14.9 m); below ~ 113 mm, nine 25.4 mm mirrors no longer pack. The architecture's map is therefore 20.4 m per standard build, 29.0 m per precision build, and 38.6 m per active alignment; larger-aperture mirrors are the identified route past all three.

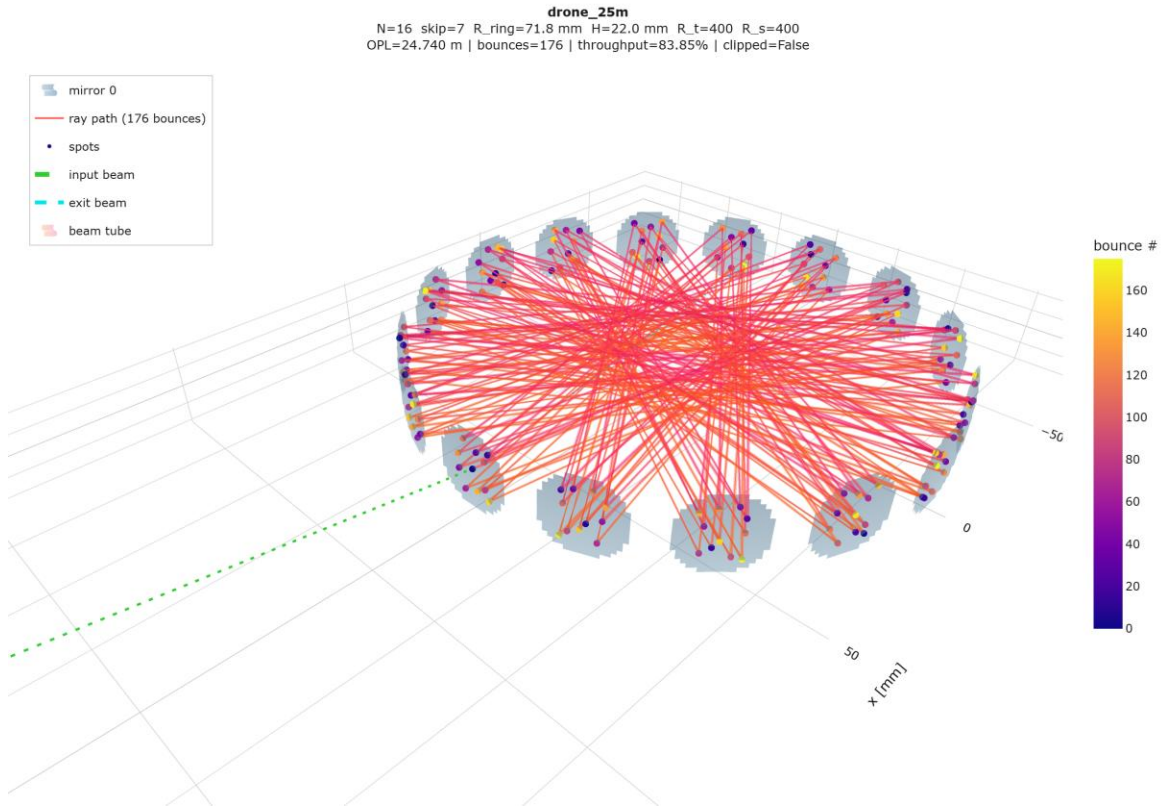


Fig. 3. The 176-chord long design (24.77 m in the same 180 mm envelope, 11 spots per mirror). At $R = 0.999$ the 32 additional reflections relative to the headline design cost only 2.8 % of signal, which is why the long-path corners are preferred when the build grade allows them.

4.2 The headline design

The 20.38 m design uses sixteen CM254-150 mirrors ($R = 300$ mm) on a ring machined to $R_{\text{ring}} = 72.155$ mm (envelope 180 mm, inner height 16 mm, sampled volume 0.26 L). The beam enters at 11.25° mean incidence, completes 144 chords of mean 141.5 mm, deposits 9 spots on each mirror with a worst-pair separation of 0.99 mm against summed beam radii of ~ 0.64 mm, keeps 4.9 mm of aperture margin, and re-

crosses the hole plane within 6 nm of the entrance point at the designed bounce; all eight intermediate visits to M0 clear the hole by at least 1.46 mm beyond the local beam radius. Mode matching sets a 0.326 mm waist 102 mm past the hole (radius at the hole 0.365 mm; hole transmission 100.0 %). Transmission is $R^{143} = 86.7\%$ at $R = 0.999$: a 10 mW DFB source delivers ≈ 8.7 mW to the detector, so the photon budget is not the limiting factor at any realistic coating.

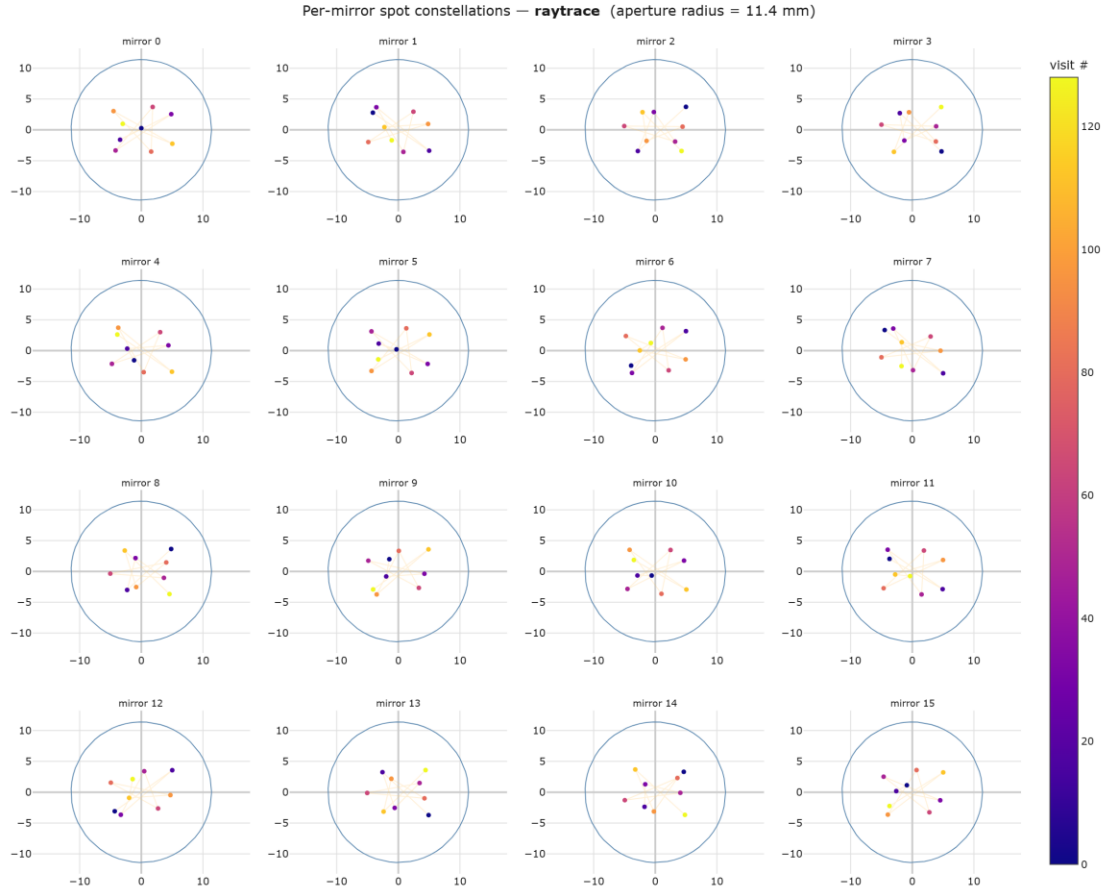


Fig. 4. Per-mirror spot constellations of the headline design (ray-traced): each of the 16 mirrors carries the same 9-point pattern at a different phase origin; the worst pair over all mirrors is the verified fringe-safety quantity. M0 additionally shows the entrance/exit hole.

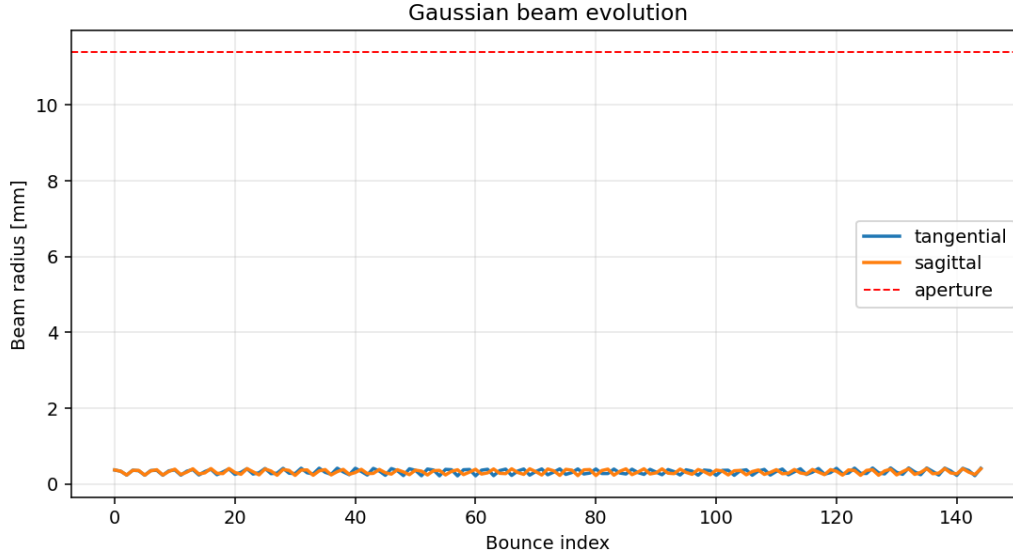


Fig. 5. Tangential and sagittal $1/e^2$ beam radii at each bounce under mode-matched injection: the beam rides the cell eigenmode at 0.21–0.42 mm for all 144 chords, an order of magnitude below the 11.4 mm clear aperture.

TMPC — as-built view (N=16, OPL=20.35 m, T=86.6%)

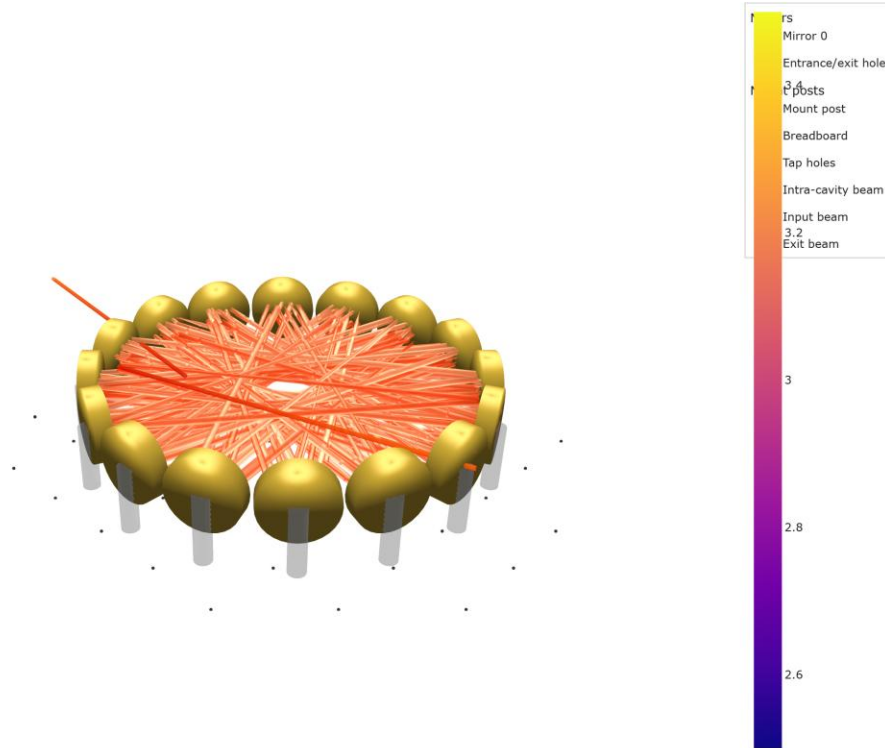


Fig. 6. As-built rendering of the headline design: sixteen 25.4 mm mirrors in a one-piece machined ring; laser collimator, mode-matching lens and detector all mount behind M0, the exit beam leaving 22.5° from the injection axis.

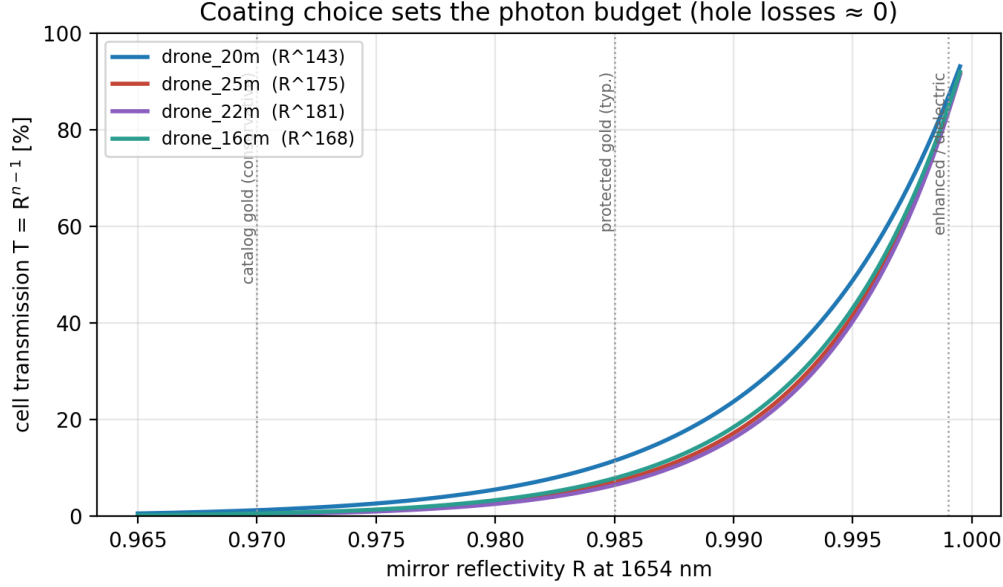


Fig. 7. Coating choice sets the photon budget: with the coupling hole rendered lossless by mode matching, transmission follows Eq. (6) for every design. Vertical guides mark conservative catalogue gold, typical protected gold, and enhanced/dielectric coatings.

4.3 Drone integration: mass and power budget

Table 2 gives a CAD-level mass estimate for the complete airborne sensor head. Mirror substrates are 25.4 mm × 6.35 mm glass (≈ 8 g each); the one-piece aluminium ring (14 mm radial section, pocketed) and two 1.5 mm lids dominate the optical-head mass and admit ~ 30 % lightweighting by skeletonising; injection optics comprise the fibre collimator, mode-matching lens and InGaAs photodiode on a common plate behind M0. With a compact DFB driver/DAQ stack the full payload is ≈ 1.0 – 1.2 kg, within the payload class of standard commercial multirotor platforms, and the 0.26 L sampled volume exchanges in well under a second at modest flow. Power is dominated by the laser TEC and electronics at a few watts; the cell itself is passive.

Table 2. Mass budget (CAD-level estimates, headline vs compact design).

item	drone_20m (Ø180)	drone_14cm (Ø141)
mirror substrates	16 × 8 g = 128 g	12 × 8 g = 96 g
Al ring (pocketed) + lids	≈ 450 – 600 g	≈ 380 – 500 g
injection optics + detector plate	≈ 80 g	≈ 80 g
optical head subtotal	≈ 0.66 – 0.81 kg	≈ 0.56 – 0.68 kg
DFB laser + driver + DAQ	≈ 0.30 – 0.40 kg	≈ 0.30 – 0.40 kg
total payload	≈ 1.0 – 1.2 kg	≈ 0.9 – 1.1 kg

5. Tolerance and engineering analysis

Each design was subjected to Monte-Carlo trials with independent per-mirror perturbations at two build grades — research (tilt $\sigma = 0.5$ mrad per axis; decentre $\sigma = 50$ μ m lateral, 100 μ m axial; curvature $\sigma = 1$ mm; ring radius $\sigma = 0.1$ mm; launch $\sigma = 50$ μ m and 0.5 mrad) and flight (0.1 mrad; 20/30 μ m; 0.5 mm; 30

μm ; 20 μm and 0.1 mrad). No trial of any tiered design clipped or lost the beam at its stated grade, and geometric quantities in Table 3 are coating-independent. One-at-a-time sensitivity identifies the dominant error source per design: mirror-pocket tilt for the $R \geq 300$ mm designs (3.3 mrad of exit drift per 0.5 mrad tilt σ , curvature error negligible), and ring radius / curvature for compact short-ROC designs (tilt negligible). The one-piece machined ring, whose pocket angles are cut in a single CNC operation, targets precisely the dominant term. Demanding build-walk-sized clearances during the search itself measurably hardens the surviving designs — the headline design's worst-case Monte-Carlo spot separation grows 30 % — turning tolerance from a post-hoc filter into a design input.

Table 3. Monte-Carlo results at research-grade tolerances (mean $\pm \sigma$; p95 in brackets).

metric	drone_20m	drone_25m	drone_16cm
trials clipping / losing beam	0/400	0/300	0/300
OPL [m]	20.35 $\pm$ 0.03	24.74 $\pm$ 0.03	20.61 $\pm$ 0.03
exit pointing drift [mrad]	4.2 $\pm$ 2.6 (9.4)	5.0 $\pm$ 3.3 (11.2)	38 $\pm$ 23 (89)
spot-pattern walk p95 [mm]	0.89	1.43	2.95
hole clearance available [mm]	1.46	3.40	1.65

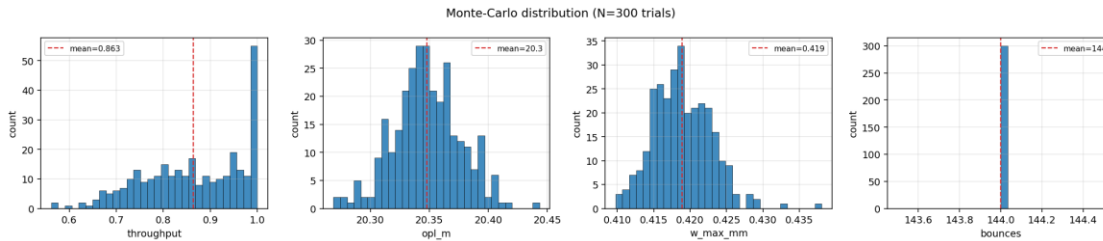


Fig. 8. Monte-Carlo distributions for the headline design at research-grade tolerances (400 trials): traversal count is invariant at 144; OPL and exit drift remain within operational bounds in every trial.

Systematic errors are compensated rather than tolerated. Catalogue mirrors carry a ± 1 % focal-length tolerance, typically biased across a delivered lot; re-optimising only the ring radius restores the complete check matrix across the entire band with a strictly linear trim — 0.72 mm per percent of curvature error for the 180 mm designs, 0.62 mm for the compact one. The assembly rule is therefore: measure the delivered curvatures, machine the ring to the interpolated radius, walk in the final micrometres with the temperature trim. Temperature acts through the same mechanism (aluminium scales R_{ring} by 23.6 ppm/K against an essentially fixed glass curvature): with the launch frozen, the headline design passes every check over ± 26 K on plain aluminium (± 20 K for the 24.8 m design, ± 8 K for short-ROC compact designs), an invar ring holds $\geq \pm 30$ K for all, and a small ring heater doubles as the closure fine-adjustment actuator. All tiered designs also pass every check up to $M^2 = 1.3$, beyond typical fibre-coupled DFB sources.

Table 4. Systematic-error compensation and robustness.

design	ROC ± 1 % $\rightarrow$ ring trim	thermal window (Al)	thermal window (invar)	M^2 tested
drone_20m	± 0.72 mm (linear)	± 26 K	$\geq \pm 30$ K	1.3 pass
drone_25m	± 0.72 mm (linear)	± 20 K	$\geq \pm 30$ K	1.3 pass
drone_16cm	± 0.62 mm (linear)	± 8 K	$\geq \pm 30$ K	1.3 pass

6. Discussion

Table 5. Position against published compact multipass cells (demonstrated values unless noted).

cell	diameter	OPL	optics	notes
Tuzson 2013 [1]	~100 mm	4.1 m	monolithic diamond-turned	first monolithic toroidal
Graf 2018 [4]	<200 g	10 m	segmented diamond-turned	interference-free
Chang 2020 [7]	100 mm	10 m (30 m theory)	ring surface	multi-layer patterns
Dong 2016 [9]	12 cm base	54.6 m	2 custom spherical	densest; critical alignment
this work, flight compact	141 mm	20.7 m	12 × catalogue \$74	81.6 % T @0.999
this work, headline	180 mm	20.4 m	16 × catalogue \$74	86.7 % T @0.999
this work, flight max	183 mm	29.0 m	12 × catalogue \$74	76.6 % T @0.999

Within the toroidal class the verified paths roughly double-to-triple the demonstrated state of the art at comparable diameter, using catalogue optics and a single machined part; the dense-pattern Herriott records [9, 10] remain denser but at custom-optics and alignment cost the one-piece ring eliminates. Three limitations bound the claims: the results are simulation-verified rather than experimentally demonstrated (the verification chain — exact tracing, independent cross-validation, Monte-Carlo tolerancing at stated build grades, systematic-error compensation — is designed to de-risk the build, and machining-level specification sheets accompany each design); coating reflectivity enters as an external parameter, quoted parametrically throughout; and surface-figure error and scatter are not yet modelled, meriting measurement on the assembled cell.

7. Conclusion

Chord-skip ring geometry, engineered dual-plane re-entrance tuned by the machined ring radius, an exact constellation criterion with build-walk-sized clearances, and mode-matched single-hole coupling together turn catalogue 1-inch mirrors into verified, tolerance-tiered 20–29 m multipass cells inside drone-compatible 133–183 mm envelopes, transmitting ~77–88 % at $R = 0.999$ on a ≈ 1 kg payload. Next steps: fabricate the headline design, measure transmission and fringe floor against the simulated budgets, and integrate a 1654 nm DFB source for airborne methane sensing.

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